

## 5. Edge Enhancement Requirement

| *An application requires highlighting edges for feature extraction.*

### (a) Interpretation of the scenario and need for edge enhancement

In applications such as object recognition, robotics vision, or automated inspection, feature extraction algorithms rely heavily on object boundaries (edges) to identify shapes, contours, and structural features. If edges in the raw image are faint, blurred, or blended with surrounding regions, downstream algorithms struggle to detect meaningful features accurately. The requirement here is to make intensity transitions at object boundaries sharper and more prominent.

### (b) Key issues in detecting edges

- Smooth or gradual intensity transitions at boundaries make edges hard to localize precisely.
- Noise in the image can create false edges or obscure genuine ones.
- Low contrast between adjacent regions weakens the gradient signal that edge detectors rely on.
- Without enhancement, feature extraction algorithms may miss real edges or produce noisy, unreliable boundary information.

### (c) Applying an appropriate method

High-pass spatial filtering is applied using a gradient-based operator such as the Sobel operator (which computes horizontal and vertical intensity gradients to highlight edges) or a Laplacian filter (which detects edges via the second derivative and can be added back to the original image for sharpening:  $g(x,y) = f(x,y) - \text{Laplacian of } f(x,y)$ ).

### (d) Justification

Gradient-based operators like Sobel directly compute the rate of intensity change in specific directions, which is exactly the information that defines an edge, making them well suited and computationally efficient for this task. The Laplacian, being isotropic (direction-independent), is useful when edges of all orientations must be enhanced uniformly. Combining Laplacian sharpening with the original image enhances edge visibility while retaining the overall image content, which is important since feature extraction algorithms need both structure and boundary information - pure edge maps alone would discard useful context.

### (e) Summary

Faint or low-contrast boundaries hinder reliable feature extraction; the key issue is a weak intensity gradient at object edges; and applying a gradient-based operator (Sobel) or Laplacian sharpening directly strengthens these transitions, producing well-defined edges suited to the application's feature extraction requirements.